

Group C Practical 2

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import argparse
import requests
import sys

WTTR_URL = "https://wttr.in/{city}?format=j1"

def fetch_weather(city, timeout=10):
    """Fetch weather JSON from wttr.in for the given city.

    Returns the parsed JSON on success. Raises requests.RequestException on network
    errors,
    and ValueError if the response isn't valid JSON or doesn't contain expected fields.
    """
    url = WTTR_URL.format(city=city)
    resp = requests.get(url, headers={"User-Agent": "weather-script/1.0"}, timeout=timeout)
    resp.raise_for_status()
    data = resp.json()

    # Basic validation
    if not isinstance(data, dict) or 'current_condition' not in data:
        raise ValueError('Unexpected response structure from wttr.in')
    return data

def format_report(data):
    """Extract and format required fields from wttr.in JSON."""
    current = data.get('current_condition', [])
    if not current:
        raise ValueError('No current condition in response')
    c = current[0]

    # Temperature (C), wind speed (kph), weatherDesc (list of dicts), weatherIconUrl (list)
    temp_c = c.get('temp_C')
    wind_kph = c.get('windspeedKmph')
    weather_desc_list = c.get('weatherDesc', [])
    weather_desc = weather_desc_list[0].get('value') if weather_desc_list else 'N/A'
    weather_icon_list = c.get('weatherIconUrl', [])
    weather_icon = weather_icon_list[0].get('value') if weather_icon_list else None

    return {
        'temperature_c': temp_c,
        'wind_kph': wind_kph,
        'description': weather_desc,
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        'icon': weather_icon,
    }

def print_report(city, rpt):
    print(f"Weather for: {city}")
    print(f"Temperature: {rpt['temperature_c']} °C")
    print(f"Wind speed: {rpt['wind_kph']} kph")
    print(f>Description: {rpt['description']}")
    if rpt['icon']:
        print(f"Icon URL: {rpt['icon']}")

def main(argv=None):
    parser = argparse.ArgumentParser(description='Fetch live weather from wttr.in')
    parser.add_argument('city', nargs='?', default='London', help='City name (default: London)')
    args = parser.parse_args(argv)

    try:
        data = fetch_weather(args.city)
        rpt = format_report(data)
        print_report(args.city, rpt)
    except requests.RequestException as e:
        print(f"Network error while fetching weather: {e}")
        return 2
    except ValueError as e:
        print(f"Error parsing weather data: {e}")
        return 3

    return 0

if __name__ == '__main__':
    sys.exit(main())

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Output

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Weather for: London
Temperature: 13 °C
Wind speed: 16 kph
Description: Partly cloudy

```